



Original Contributions

Longevity-promoting effects of 4-hydroxy-E-globularinin in *Caenorhabditis elegans*Virendra Shukla^a, Deepti Yadav^b, Suresh C. Phulara^a, M.M. Gupta^b, Shilpi K. Saikia^a, Rakesh Pandey^{a,*}^a Microbial Technology and Nematology, CSIR–Central Institute of Medicinal and Aromatic Plants, Lucknow 226015, India^b Analytical Chemistry Division, CSIR–Central Institute of Medicinal and Aromatic Plants, Lucknow 226015, India

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ABSTRACT

In modern times, there has been a major increase in the use of plants or herbal constituents for the prevention of age-related disorders. 4-Hydroxy-E-globularinin (4-HEG) is an iridoid and a major component of *Premna integrifolia*. This investigation represents a breakthrough in geriatrics by showing the longevity-promoting activity of 4-HEG in the animal model *Caenorhabditis elegans*. 4-HEG (20 μ M) enhanced the mean life span of worms by over 18.8% under normal culture conditions and also enhanced their survival under oxidative stress. The longevity-promoting activity was associated with reduced reactive oxygen species (ROS) levels and fat accumulation in the worms. Gene-specific mutant studies verified the role of ROS detoxification pathways and simultaneous nuclear translocation of DAF-16 in the 4-HEG-mediated effects. Quantitative real-time PCR estimations and observations of transcriptional reporters indicated that 4-HEG was able to upregulate stress-inducible genes, viz., *hsp-16.2* and *sod-3*. Thus, 4-HEG may serve as a lead compound of plant origin for the development of important nutraceuticals superseding the aging process.

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Aging and age-related diseases have an ever-increasing economic impact on world health. As a result there is an intense interest in developing medicines to delay or prevent the decline in tissue function with age. Reactive oxygen species (ROS) are chemically active molecules generated during cellular metabolism and by various stresses [1]. ROS are able to react with biomolecules and damage physiologically vital macromolecules such as DNA, lipids, and proteins, leading to numerous age-related diseases such as cancer, arthritis, cataract, osteoporosis, type 2 diabetes, hypertension, and Alzheimer disease [2,3]. Because the majority of the world population (approximately 60–80%) still relies on traditional systems of medicine for the cure of common diseases [4], the search for drugs of plant origin with antioxidant activities, which scavenge ROS, has become a central focus of antiaging research.

Premna integrifolia Linn. (syn: *Premna serratifolia*), commonly known as “agnimantha,” is one of the important constituents of the famous herbal remedy “dashmula,” a 10-herb formulation commonly used in the Indian system of medicine. The plant is useful in the treatment of skin diseases, arthritis, gonorrhea, rheumatism, anorexia, and jaundice. Many pathological activities of *P. integrifolia*, such as antirheumatic, carminative, galactogenic, bechic, febrifuge, stomachic, and anti-inflammatory activities

have been reported [5]. The decoction of roots is also used to treat gonorrhea. Phytochemical studies on this plant reveal the presence of β -sitosterol, polyisoprenoid, spermine alkaloids, aphilandrine, premnin, and iridoids [6,7].

Iridoids are a group of natural constituents with a monoterpane cyclic ring, found in a wide variety of plants. They have a defensive role in certain plant species [8] and also have a wide range of pharmacological effects in animals such as treatment of hepatic dysfunction, stimulation of bile acid excretion, and anti-tumor and anti-inflammatory activities [9–11]. The antioxidant activity of iridoids has been reported by several researchers [12,13] but the antiaging activity of the compounds belonging to the iridoid group is yet not explored, and much effort is required to detect the possible role and mode of action of these compounds. In addressing this aspect, this study was designed to isolate and characterize an iridoid compound, 4-hydroxy-E-globularinin (4-HEG), from *P. integrifolia*. The purified compound was studied for its antiaging potential using the animal model *Caenorhabditis elegans*. Since the discovery that a single gene mutation could produce a substantial increase in its life span, the nematode *C. elegans* has emerged as an important model for aging research, as most of the *C. elegans* genes have human orthologues and the loss of function of the equivalent genes can extend the life span of mammals [14]. This suggests that mechanisms found to influence aging in *C. elegans* are likely to have a conserved role in regulating longevity in humans. This study will help to explore and authenticate such plant-derived molecules

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in the development of useful prolongevity pharmaceuticals and also has the potential to be of great economic benefit to society.

Materials and methods

Plant material

Stem bark of *P. integrifolia* was collected from Tambarum, Chennai, India. Plant identification was done by Professor P. Jayaraman, Director, National Institute of Herbal Science, Plant Anatomy Research Centre, West Tambaram, Chennai, India. A voucher specimen (No. 22706) has been deposited in the herbarium of the institute.

Apparatus for isolation and identification

The 300-MHz NMR spectra were recorded with tetramethyl silane as internal standard on a Bruker Avance instrument. ^{13}C NMR and DEPT spectra were recorded at 75 MHz. The DEPT experiments were used to resolve multiplicities of carbon atoms. Chemical shifts are given in parts per million. COSY, HSQC, and HMBC were performed using standard Bruker pulse programs. Infrared spectra were obtained on a Nicolet 380 FTIR spectrophotometer. Mass spectra were measured on an API 3000 triple-quadrupole mass spectrometer (Applied Biosystems/MDS SCIEX). Optical rotations were measured on a Horiba polarimeter (SEPA-300). Column chromatography was executed using silica gel (60–120 mesh; Merck, India). Preparative HPLC (Shimadzu, Japan) consisting of a column (Supelcosil LC-18, 21.2×250 mm, $12 \mu\text{m}$), pumps LC-8 A, and PDA detector was used for isolation and purification of reference compounds [15].

Extraction and isolation of 4-HEG from *P. integrifolia*

Air-dried and finely powered stem bark of the plant (900.0 g) was extracted with methanol ($2 \text{ L} \times 24 \text{ h}$) at room temperature; the process was repeated five times and combined extracts were evaporated under reduced pressure to obtain 53.0 g residue. Methanolic extract was suspended in water and then partitioned with hexane, chloroform, and butanol, successively. The *n*-hexane-partitioned fractions were combined and concentrated in vacuo to give a residue (2.7 g). Similarly, chloroform and butanol fractions were also concentrated to give residues of 1.5 and 23.0 g, respectively. The butanol extract was chromatographed over a silica gel column (60–120 mesh, $100 \times 5 \text{ cm}$), eluted with CHCl_3 and increasing polarity with MeOH, and was fractionated into 10 fractions: fr. B1–B10 (each 4.5 L, *n*-hexane:EtOAc 1:0, 19:1, 23:2, 9:1, 17:3, 4:1, 3:1, 7:3, 1:1, EtOAc). Fr. B4 (1.8 g) was chromatographed to a reverse-phase preparative HPLC column (RP-C18, MeOH:H₂O 70:30, 15 ml/min), which yielded compound 4-HEG (75 mg) (t_R 7.1). The structure elucidation of compound (Fig. 1) was performed with the help of UV, ^1H NMR,

^{13}C NMR, 2D experiments (HSQC, HMBC, COSY, and NOESY), and mass analysis (ESI–MS) and confirmed finally by comparison with reported data [16]. The ^1H and ^{13}C NMR data of marker compounds are presented in Table 1.

The isolated compound was then dissolved in double-distilled water to prepare stock solutions of 20, 2, and 0.2 mM.

C. elegans strains

N2 Bristol (wild type); TK22, *mev-1(kn1)*; GR1307, *daf-16(mgDf50)*; VC-199, *sir-2.1(ok434)*; TJ356, (*zls356*), *daf-16::gfp*; CF1553, *mul84(pAD76(sod-3::gfp))*; and CL2070, *dvl570(hsp-16.2::gfp; rol-6(su1006))* strains of *C. elegans* and *Escherichia coli* OP50 were kindly provided by the *Caenorhabditis* Genetics Center (CGC), University of Minnesota (Minneapolis, MN, USA). The strains were routinely cultured and maintained at 20°C on nematode growth medium (NGM) seeded with *E. coli* OP50 bacteria [17].

Life span assay

Age-synchronized N2 worms were used for life span assay. Isolated eggs were allowed to hatch on NGM plates previously spotted with or without various concentrations of 4-HEG, viz., 200, 20, and $2 \mu\text{M}$ until L4 stage. Thirty to forty L4 molts were then transferred to NGM plates previously spotted with corresponding concentration and $50 \mu\text{M}$ 5-fluorodeoxyuridine (FUDR; Sigma, St. Louis, MO, USA) to block progeny development. Worms were then observed daily for survival and transferred to fresh plates after every 3–4 days to avoid contamination and to ensure the presence of the compound throughout the experiment [18].

Stress resistance assay

To assess effect of 4-HEG on the resistance of worms against oxidative stress, paraquat (Sigma), which is an intracellular ROS generator, was used. Wild-type N2 worms were synchronized on NGM plates as described for life span assay. Adult day 2 worms

Table 1

^1H and ^{13}C NMR shifts in δ for compound 4-HEG (in CDCl_3).

Position	$\delta \text{ H (m, J in Hz)}$	$\delta \text{ C}$
1	5.30 (1 H, d, 6.3)	95.3
2	—	—
3	6.35 (1 H, dd, 6.0, 2.1)	140.7
4	5.10 (1 H, m)	105.6
5	2.7 (1 H, m)	38.1
6	3.95 (1 H, m)	79.0
7	3.87 (1 H, d, 4)	79.0
8	—	81.7
9	2.39 (1 H, dd, 10.2, 6.3)	44.0
10 A	4.35 (1 H, d, 11.3)	68.1
10 B	4.52 (1 H, d, 11.3)	—
1'	4.71 (1 H, d, 7.8)	100.0
2'	3.21 (1 H, dd, 9.0, 8.0)	74.8
3'	3.42 (1 H, m)	77.9
4'	3.15 (1 H, m)	71.2
5'	3.32 (1 H, m)	78.2
6'	3.82 (2 H, dd, 12.0, 2.0)	61.8
1''	—	127.2
2'',6''	7.49 (1 H, d, 9.0)	132.3
3'',5''	6.82 (1 H, d, 9.0)	115.8
4''	—	161.2
7''	7.68 (1 H, dd, 16.2, 4.2)	145.9
8''	6.39 (1 H, d, 15.9)	114.2
9''	—	169.8

^1H NMR (300 MHz) and ^{13}C NMR (75 MHz) recorded in MeOD; chemical shifts are expressed in ppm and J values in Hz.

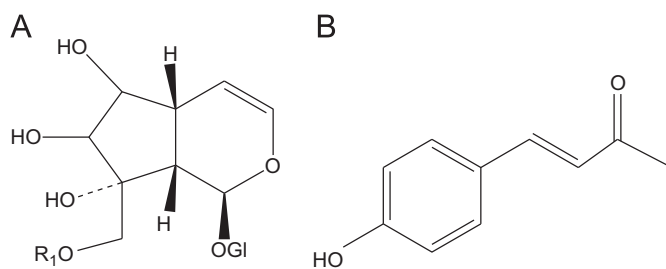


Fig. 1. (A) Structure of 4-hydroxy-E-globularinin. (B) Structure of R1, which is a part of 4-HEG.

were transferred to 10 mM paraquat and assayed for survival after 72 h of continuous exposure [19].

Pharyngeal pumping assay

To investigate the effect of 4-HEG on pharyngeal pumping of the worms, the movement of the pharynx terminal bulb was recorded. Worms were grown on NGM plates and treated with or without various concentrations of 4-HEG from L1 stage. Pumping records were made for 20-s intervals on adults at day 0 and day 5. The experiment was performed at room temperature and worms were scored on a bacterial lawn [20].

Chemotaxis assay

Chemotaxis assay was performed according to Bargmann et al. [21], using adult day 5 worms with slight modifications. Briefly, treated and untreated worms were washed in M9 buffer, and 90 worms were placed in the center of agar plates containing 10 μ l of attractant on one side (1 M sodium acetate) and 10 μ l of water on the other side. Sodium azide (10 μ l of 1 M) was added on both sides to paralyze the worms if they happened to venture in the direction opposite the attractant. Worms were counted after 90 min and chemotaxis index (CI) was calculated as

$$CI = A - B / A + B,$$

where *A* is the number of worms at the attractant location, *B* is the number of worms at the control location, and *A* + *B* is the total number of worms in the plate. The experiments were repeated independently three times.

Green fluorescent protein (GFP) visualization and quantification

The intracellular localization of DAF-16 was observed using the TJ356 mutant, in which GFP is fused with the *daf-16* gene. The genes downstream of *daf-16*, viz., *hsp-16.2* and *sod-3*, were observed through the inducible GFP-reporter-tagged CL2070 and CF1553 *C. elegans* strains. Age-synchronized L4 worms were transferred to NGM plates treated or not with 4-HEG and kept at 20 °C for 48 h. Untreated worms that underwent a short thermal stress of 37 °C for 20 min were used as positive controls. Worms were divided into three groups (nuclear, cytoplasmic, and both) on the basis of DAF-16::GFP fusion protein localization. The nuclear category included the worms showing fluorescence in the nuclei region, and the worms without nuclear fluorescence were categorized as cytoplasmic. Worms that did not fit in any category were classified as both [22,23]. Worms were then mounted on a 2% agarose pad and visualized under a microscope (Carl Zeiss AxioScope A1). Photomicrographs for strain CL2070 were taken after exposure of the worms to 37 °C for 2 h and their substantial recovery at 20 °C for 4 h, whereas the photographs for strain CF1553 were taken directly at day 3 of the treatment [24]. For GFP quantification images were analyzed by ImageJ software (NIH).

Nile red staining

Nile red staining was done as previously described by Singh et al. [25], with minor modifications of using solid medium instead of liquid medium. To investigate fat accumulation worms were grown on NGM plates supplemented with (*n*=100) or without (*n*=94) 20 μ M 4-HEG from L1. On adult day 1 worms were transferred to corresponding treatment plates cosupplemented with Nile red (75 ng/ml) for 3 days. Worms were mounted on a 2% agarose pad and visualized with a fluorescence microscope (Carl Zeiss AxioScope A1) at 10 $\times$ magnification.

ROS detection assay

Intracellular ROS level was determined using a nonfluorescent dye, dichlorodihydrofluorescein diacetate (H₂DCF-DA; Sigma). Worms were synchronized on NGM plates as described earlier. At day 2 of adulthood 30 age-synchronized worms were collected in 300 μ l of 0.1% phosphate-buffered saline with Tween 20 (PBST). Worms were washed three times with PBST and the lysate was prepared by equally timed homogenization and sonication (Branson Sonifier 250; VWR Scientific, Suwanee, GA, USA). The homogenized samples were transferred to 96-well plates and, before being read, 15 μ l of 10 mM H₂DCF-DA was added to each well. Fluorescence readings were taken every 20 min for 2 h and 30 min at 37 °C using a fluorescence microplate reader (Spectra-max M₂; Molecular Devices). Excitation was read at 485 nm and emission was read at 535 nm [3].

Gene quantification through real-time PCR

N2 wild-type worms were grown under standard laboratory conditions at 20 °C with 20 μ M 4-HEG-treated and untreated plates. Total RNA was isolated using the guanidinium acid–phenol–chloroform method [26] and TRIzol reagent (Invitrogen). First-strand cDNA synthesis was performed with an equal amount of RNA using the Thermoscript real-time PCR kit (Invitrogen) as per the kit's instructions. The genes observed through real-time PCR were as follows: *hsp-16.2*, *sod-3*, and *gst-7*. The housekeeping gene *act-1* was used as an internal control. Primer sequences are available on request. mRNA expression was quantified using the SYBR green detection method on an Applied Biosystems 7900HT fast real-time PCR system. Relative quantification for the expressed genes was done using the comparative C_T ($\Delta\Delta C_T$) method [27].

Statistical analysis

Significant differences between the life spans of treated and control worms under both normal and stress conditions were determined using the Kaplan–Meier survival assay in MedCalc software. Data for other than life span were statistically analyzed using ANOVA in ASSISTAT statistical assistance software. Differences between the data were considered significant at *P* ≤ 0.05.

Results

4-HEG increased the mean life span of wild-type N2 worms

To investigate whether 4-HEG supplementation has any effect on life span of wild-type *C. elegans*, the age-synchronized N2 worms were subjected to various concentrations of 4-HEG, viz., 200, 20, and 2 μ M at 20 °C. Worms were observed daily for survival by the touch-provoked method [28] and the survival of 4-HEG-supplemented worms was compared with that of untreated worms. In this investigation all the tested concentrations increased the mean life span of wild-type worms at 20 °C (Table 2). The 20 μ M concentration of 4-HEG most significantly (*P* < 0.0001) increased the mean life span of wild-type worms, by 18.8% compared to untreated control worms (Table 2 and Fig. 2 A). An increase in maximum and minimum life span was also found in 20 μ M-treated worms.

4-HEG reduced oxidative stress in *C. elegans*

It has been already established that increased longevity is closely associated with improved survival under stress conditions [29]. To investigate the longevity-promoting ability of 4-HEG under

Table 2Effect of 4-HEG on the life span of *C. elegans* at 20 °C.

Strain	Treatment (μM)	Mean life span ± SE	No. of worms	Change (%)	P value	Max life span ± SE	Min life span ± SE
N2	Control	18.6 ± 0.22	234			25.7 ± 0.25	11.0 ± 0.40
N2	4-HEG (200)	19.8 ± 0.45	228	6.4	0.0056	26.2 ± 0.62	12.2 ± 0.48
N2	4-HEG (20)	22.1 ± 0.26	237	18.8	< 0.0001	28.5 ± 0.50	16.5 ± 0.64
N2	4-HEG (2)	20.1 ± 0.31	174	8.06	0.0001	27.0 ± 0.50	12.3 ± 0.28
<i>mev-1 (kn-1)</i>	Control	14.3 ± 0.35	225			20.0 ± 0.57	6.7 ± 0.33
<i>mev-1 (kn-1)</i>	4-HEG (20)	14.0 ± 0.39	249	−2.09	0.11	19.0 ± 0.58	7.3 ± 0.67
<i>daf-16 (mgDf50)</i>	Control	15.4 ± 0.22	273			20.7 ± 0.67	8.3 ± 0.88
<i>daf-16 (mgDf50)</i>	4-HEG (20)	15.5 ± 0.12	270	0.65	0.43	21.0 ± 0.81	7.7 ± 0.40
<i>sir-2.1 (ok434)</i>	Control	16.9 ± 0.64	294			23 ± 0.40	9.0 ± 0.40
<i>sir-2.1 (ok434)</i>	4-HEG (20)	20.3 ± 0.39	222	20.1	< 0.0001	25 ± 0.57	11.7 ± 0.88

The life span assay was done with wild-type N2 and *mev-1(kn-1)*, *daf-16(mgDf50)*, and *sir-2.1(ok434)* mutants at 20 °C. Worms were treated with various test concentrations of 4-HEG from L1 stage until the completion of the assay. The mean life span was calculated as the average number of days the worms survived in each test concentration.

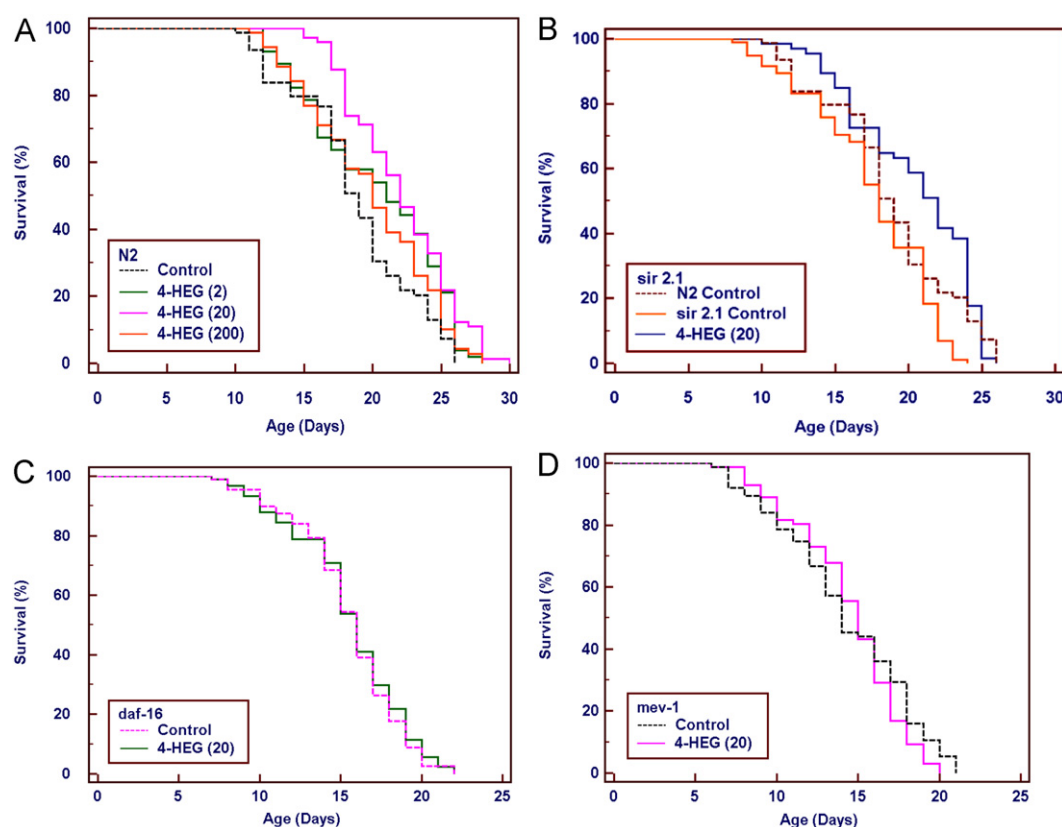


Fig. 2. Effect of 4-HEG on life span of *C. elegans*. Worms were synchronized on NGM plates spotted with various test concentrations of 4-HEG (2, 20, and 200 μM) on an OP50 lawn and allowed to grow until L4 stage. L4 molts were then transferred to fresh NGM plates of corresponding test concentration with 50 μM FUDR. Worms were scored daily for survival. The 20 μM test concentration showed the most significant increase in life span of (A) wild-type N2 worms ($P < 0.0001$) and (B) the *sir-2.1* mutant ($P < 0.0001$), whereas no increase in mean life span was observed in (C) *daf-16* or (D) *mev-1* mutants. Survival plot was drawn by Kaplan–Meier survival assay.

oxidative stress, wild-type worms were exposed to 10 mM paraquat. Worms were synchronized on NGM plates with various concentrations of 4-HEG along with controls. On adult day 2 worms were transferred to NGM plates containing 10 mM paraquat and the survival of worms was recorded after 72 h. It was observed that worms pretreated with all the test concentrations of 4-HEG were more resistant to paraquat than the untreated control worms (Fig. 3 and Supplementary Table 1). The highest survival was recorded in worms pretreated with 20 μM 4-HEG ($P < 0.01$).

4-HEG reduced free radical levels and fat accumulation in *C. elegans*

To understand the potential effects of 4-HEG under oxidative stress, the intracellular free radical level of wild-type worms was

determined after 4-HEG supplementation. It was observed that 4-HEG supplementation in all tests reduced ROS generation in wild-type worms (Fig. 4A). The lowest level of ROS production (43%) was observed in 20 μM 4-HEG-pretreated worms ($P < 0.001$).

In *C. elegans*, increased fat accumulation is also associated with accelerated aging and higher ROS levels [25,30]. To investigate the possible effect of 4-HEG on fat accumulation in *C. elegans*, wild-type N2 worms pretreated with 20 μM 4-HEG were stained with Nile red. However, it has been well reported that Nile red staining intensities do not necessarily correlate with fat levels in live worms [31]. After 3 days continuous exposure to Nile red, fluorescence images of the worms were captured (10 × original magnification). Fluorescence intensity was observed as pixel per unit area using ImageJ software. It was observed that 4-HEG

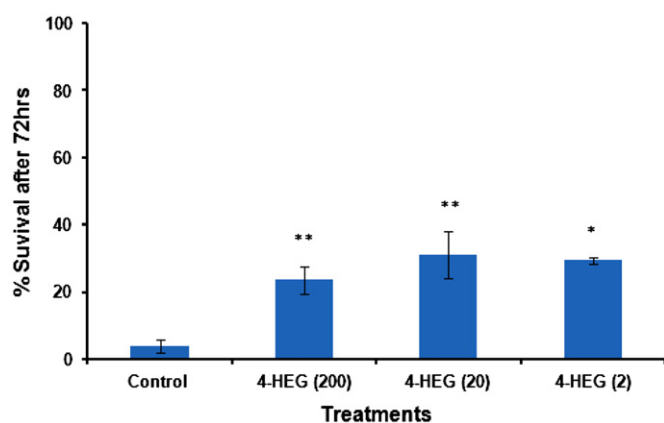


Fig. 3. Effect of 4-HEG on paraquat-induced oxidative stress on wild-type N2 worms. Worms were synchronized on NGM plates supplemented with (200 μ M, $n=114$; 20 μ M, $n=96$; 2 μ M, $n=102$) or without (control, $n=106$) 4-HEG. On adult day 2 the worms were transferred to paraquat (10 mM)-treated plates and survival of the worms was measured after 72 h. All test concentrations significantly reduced the paraquat sensitivity in pretreated worms compared to untreated control worms. Error bars represent the standard error of the mean. Statistically significant at * $P < 0.05$, ** $P < 0.01$.

treatment reduced fat accumulation in wild-type worms compared to untreated control worms (Fig. 4B and C). ANOVA was performed to observe the significant difference between the fat accumulations. The results revealed that 20 μ M 4-HEG supplementation significantly reduced fat accumulation inside wild-type worms ($P < 0.001$ and Supplementary Table 2).

4-HEG enhanced the pharyngeal pumping of *C. elegans*

The speed of pharynx contraction or pumping is one of the parameters of aging in *C. elegans* and an alteration in pumping can induce dietary restriction (DR)-like effects in worms [32]. To investigate this possibility the movement of the pharynx bulb was recorded for 20 s on adult day 0 and day 5. We did not observe any decrease in pharyngeal pumping on day 0 of adulthood, whereas a significant increase was found in 2 μ M-treated worms ($P < 0.05$). At adult day 5 pumping was higher in 20 and 2 μ M-treated worms than in the untreated worms ($P < 0.01$, $P < 0.001$, respectively; Fig. 5A and Supplementary Table 3).

4-HEG increased the chemotaxis index of the worms

To investigate the effect of 4-HEG on age-dependent physiological behavior the CI was determined, which is a measure of the learning behavior and motor responses to a sensory stimulus in *C. elegans* [21]. Adult day 5 wild-type worms were used to investigate the effect of 4-HEG on CI. Sodium acetate was used as an attractant and sodium azide was used to paralyze the worms to prevent their movement in the opposite direction. We observed an increase in CI in 2 and 20 μ M-treated worms ($P < 0.01$ and $P < 0.001$, respectively), whereas the 200 μ M concentration was unable to produce any such effect (Fig. 5B).

Life-span extension due to 4-HEG required the insulin signaling and ROS detoxification pathways

In *C. elegans* genes responsible for life-span extension have been already identified, which empowers the use of its mutants for identification of pathways modulated by various drugs. In this experiment attempts were made to identify the possible mechanism of action of 4-HEG, utilizing various worm mutants. In *C. elegans*, *sir-2.1* encodes a histone deacetylase-like protein that brings metabolic

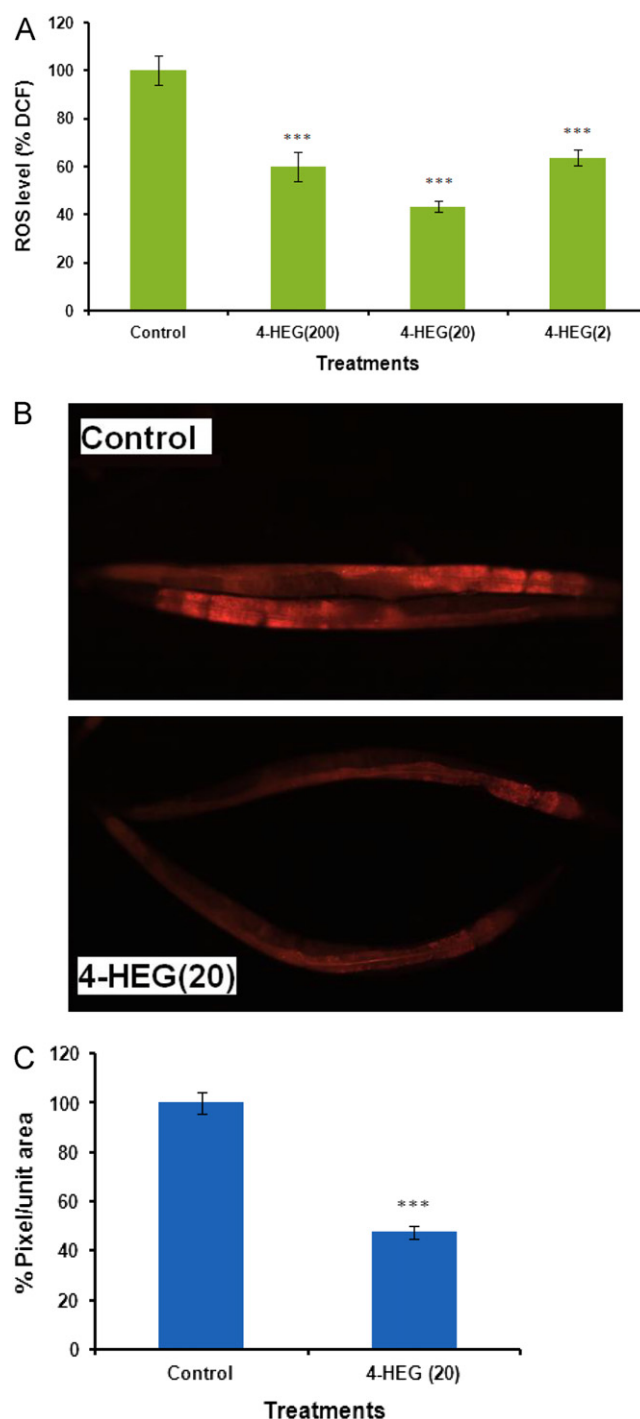


Fig. 4. Effect of 4-HEG on ROS and fat accumulation in *C. elegans*. To measure intracellular ROS levels age-synchronized wild-type N2 worms were treated with various concentrations of 4-HEG from the L1 stage. On adult day 2 the worms were homogenized and ROS levels were detected using a 96-well microplate reader at 37 °C. The graph was plotted as relative change in ROS compared to control at 100%. (B) To investigate fat accumulation worms were grown on NGM plates supplemented with ($n=100$) or without ($n=94$) 20 μ M 4-HEG from L1. On adult day 1 the worms were transferred to corresponding treatment plates cosupplemented with Nile red (75 ng/ml) for 3 days. Images of worms were captured using a 10 $\times$ objective. (C) Pixel/unit area of Nile red-stained worms was quantified by ImageJ (NIH) software and significant difference was determined using ANOVA. Error bars represent the standard error of the mean. Statistically significant at *** $P < 0.001$.

status together with life span [33,34]. 4-HEG supplementation increased the life span of *sir-2.1* mutants significantly by 20.1% ($P < 0.0001$; Fig. 2B, Table 2). The *C. elegans* mutant *daf-16(mgDf50)*

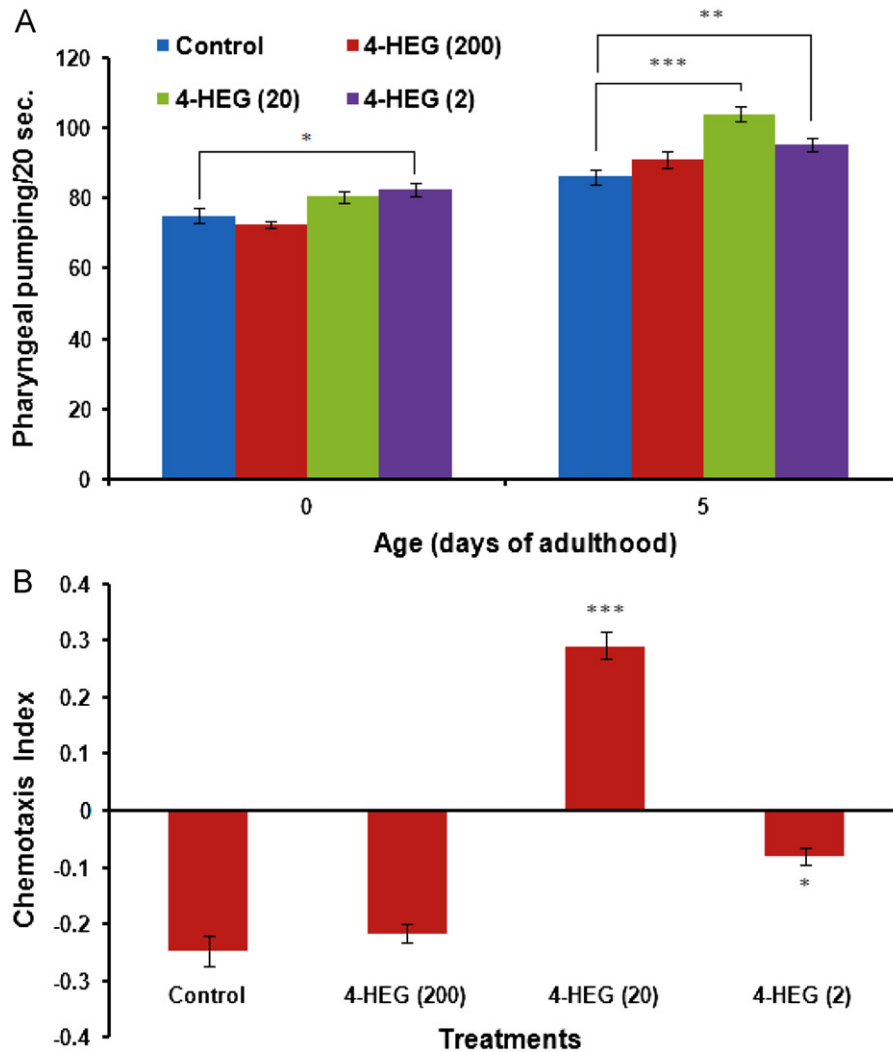


Fig. 5. Effect of 4-HEG on the pharyngeal pumping and CI of wild-type worms. (A) Pharyngeal pumping of worms treated with various concentrations of 4-HEG was recorded for 20 s and compared with untreated control worms. On adult day 5, 20 and 2 μ M concentrations enhanced the movement of the pharynx bulb significantly. 20 worms per experimental condition were scored and the experiment was repeated three times. (B) For CI wild-type *C. elegans* were synchronized on NGM plates with (200 μ M, $n=92$; 20 μ M, $n=90$; 2 μ M, $n=96$) or without (control, $n=90$) 4-HEG. CI was determined using adult day 5 worms. Sodium acetate was used as attractant and sodium azide was applied to paralyze the worms. After 90 min the worms were scored and CI was calculated. All the test concentrations enhanced the CI significantly. Error bars represent the standard error of the mean. Statistically significant at $*P < 0.05$, $**P < 0.01$, $***P < 0.001$.

has a null mutation in its gene encoding DAF-16, a major transcription factor of the IIS pathway, conferring stress tolerance and longevity [12]. It was observed that 4-HEG supplementation failed to increase the life span of this mutant (Fig. 2C; Table 2). *mev-1* mutants have a mutation in the cytochrome *b* subunit of mitochondrial complex II and are thus short-lived [35]. Overproduction of superoxide and increased oxidative stress in these mutants results from genetic changes along with accelerated aging and reduced life span. Similarly, no increase in life span was observed in *mev-1(kn-1)* mutants after 4-HEG supplementation (Table 2 and Fig. 2D). These results indicate that 4-HEG reduces oxidative stress and also requires an endogenous detoxification pathway for reducing the same.

Further, the genetic requirement of insulin signaling for the effects mediated by 4-HEG was also examined using the TJ356 mutant carrying a *daf-16::gfp* transgene. To investigate the nuclear translocation, worms were treated with 20 μ M 4-HEG for 48 h and then visualized under a fluorescence microscope. A small heat shock at 37 $^{\circ}$ C for 20 min resulted in approximately 100% nuclear translocation of DAF-16. It was observed that 4-HEG supplementation was able to translocate 45% of DAF-16 into the nucleus (Fig. 6).

4-HEG enhanced expression of HSP-16.2 and SOD-3 in transgenic *C. elegans*

The transgenic lines of heat shock protein *hsp-16.2::gfp* and superoxide dismutases *sod-3::gfp* were studied to evaluate the stress-related molecular responses to 4-HEG treatment in *C. elegans*. HSP-16.2 is an anti-stress reporter protein, playing a key role in longevity, whereas superoxide dismutase (*sod-3*) is an antioxidant enzyme, equally important as a regulator of life span and stress resistance. Transgenic strains of the nematode, *hsp-16.2::gfp* and *sod-3::gfp*, were treated with 20 μ M 4-HEG and *hsp-16.2::gfp* was exposed to thermal stress. The expression of *hsp-16.2* and *sod-3* was examined by observing the intensity of green fluorescent protein. In this experiment the quantification of fluorescence intensity of *hsp-16.2::gfp* and *sod-3::gfp* using ImageJ software was done. 4-HEG-treated worms had significantly increased fluorescence intensity of HSP-16.2 and SOD-3 proteins compared to untreated control worms. The mean fluorescence (GFP) intensity of *hsp-16.2* was 5.8 ± 0.52 arbitrary units in control worms and 8.5 ± 0.75 arbitrary units in 20 μ M 4-HEG-treated subjects (Supplementary Figs. 2A, B, and E). A 1.5-fold

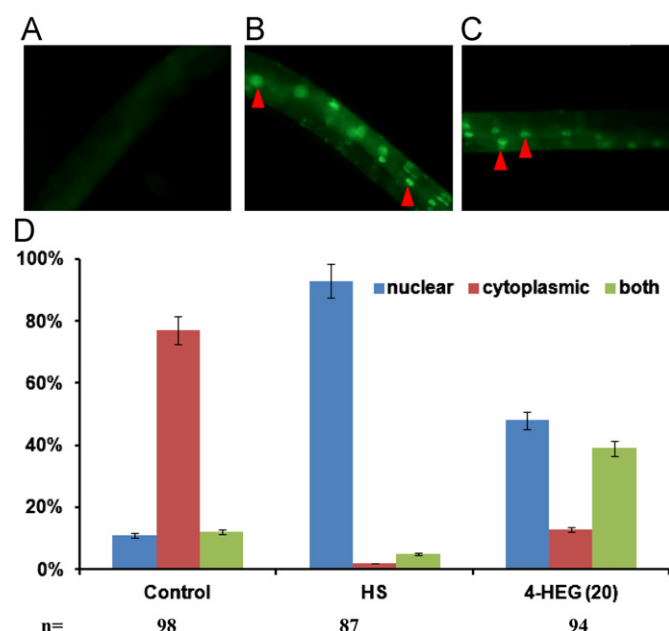


Fig. 6. Effect of 4-HEG on nuclear localization of DAF-16. (A) Cytosolic DAF-16 localization (control worm). (B and C) Nuclear DAF-16 translocation in a heat shock (HS)- and 20 μ M 4-HEG-treated worm. (D) Quantification of the DAF-16::GFP localization. Age-synchronized L4 TJ356 worms were continuously exposed to 20 μ M 4-HEG for 48 h. Worms that underwent a small heat shock at 37 °C for 20 min were used as positive controls. Supplementation of 4-HEG resulted in nuclear translocation of DAF-16. Worms were mounted on a 2% agarose pad on a glass slide and the subcellular (nuclear, cytoplasmic, and both (nuclear+cytoplasmic)) localization of DAF-16 was analyzed under a fluorescence microscope. The results are presented as means $\pm$ SEM of the percentages of DAF-16 localization. The experiment was repeated three times and images were captured using a 40 $\times$ objective. n, number of worms imaged.

increment ($P=0.03$) in HSP-16.2 was observed in 4-HEG-exposed worms. Consistent with *hsp-16.2*, in *sod-3* mutants the mean fluorescence (GFP) intensity was higher in 4-HEG-treated worms compared to control worms (Supplementary Figs. 2C, D, and E). A significant ($P=0.05$) 1.4-fold upregulation of SOD-3 level was observed in the 4-HEG-treated worms.

4-HEG regulates the mRNA expression of *daf-16*-dependent stress-inducible genes

DAF-16 is a major transcription factor that regulates the life span, stress resistance, and antioxidant gene expression through nuclear localization. Quantitative real-time PCR experiments were performed to investigate whether 4-HEG could regulate the expression of genes downstream of *daf-16*. Our results showed that 4-HEG significantly upregulated the mRNA expression of *hsp-16.2* ($P<0.001$) and *sod-3* ($P=0.003$; Fig. 7), but did not affect the expression of *gst-7* ($P>0.5$). The upregulation of *hsp-16.2* and *sod-3* is consistent with the result for HSP-16.2::GFP and SOD-3::GFP reporters (Supplementary Fig. 2). Our results revealed that 4-HEG could be promoting the life span and stress resistance of *C. elegans* through *daf-16*.

Discussion

This study demonstrates the isolation and longevity-promoting activity of iridoid 4-HEG from an important medicinal plant, *P. integrifolia*. The study is the first report of the antiaging activity of one of iridoids in the most promising animal model for aging research, i.e., *C. elegans*. Previously with this model many

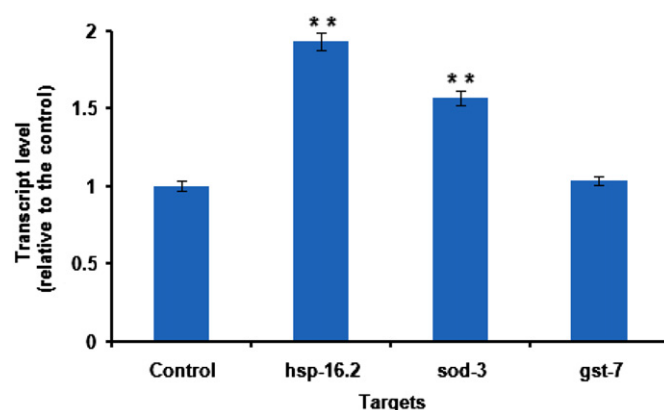


Fig. 7. Effect of 4-HEG on the expression of the stress-inducible genes *hsp-16.2*, *sod-3*, and *gst-7* in wild-type *C. elegans*. Expression levels of genes relative to actin-1 were determined by real-time PCR in worms treated with 20 μ M 4-HEG. Actin-1 was taken as an internal control. Error bars represent the standard error of the mean. Statistically significant at ** $P<0.01$.

successful studies have been made using natural compounds such as blueberry polyphenols [19], resveratrol [36], epigallocatechin gallate [20], reserpine [37], quercetin [38], curcumin [30], and diallyl trisulfide [32], among others. Antioxidants are the compounds that are able to reduce ROS through their ability to bind with free radicals. Iridoids are well recognized for their antioxidant properties [12,13]. These reports encouraged us to seek the longevity-promoting activity of 4-HEG using *C. elegans* and contribute the findings to the ongoing search for a novel compound promising extended life span.

These experiments demonstrated that 4-HEG at its 20 μ M concentration significantly increased the mean life span of wild-type N2 worms under normal culture conditions at 20 °C. There is a correlation between stress resistance and longevity [28,29]. Like in previous studies [37–39], in our experiment increased life span was also associated with stress tolerance, as 4-HEG-treated worms showed increased resistance to paraquat-induced oxidative stress. Survival curves have revealed that 4-HEG supplementation enhanced the worms' survival under oxidative stress (Supplementary Fig. 1). Maximum survival was recorded in 20 μ M 4-HEG-supplemented NGM plates after 72 h. According to the free radical theory of aging, it is evident that increasing oxidative damage decreases life span and vice versa. Age-related diseases are frequently accompanied by oxidative stress. Living cells manage the oxidative damage (ROS) through numerous antioxidant enzymes, viz. superoxide dismutase, catalase, and glutathione peroxidase [40,41]. Free radicals play an important role in several biological pathologies and contribute to the formation of ROS, which ultimately damage biomolecules. Our results demonstrate that the intracellular ROS level was lower in 4-HEG-treated worms (Fig. 4A). Similar to previous findings [19,22,40], the low level of ROS can be assumed as a responsible factor for increased resistance against oxidative stress and the enhancement of life span under normal culture conditions. In another experiment the effects of 4-HEG on fat storage in worms were examined. It was found that fat level was reduced in 4-HEG-supplemented worms compared to the untreated worms (Fig. 4B and C). Statistical analysis also revealed that fat accumulation was 52.5% less in 4-HEG-supplemented worms than in the untreated worms ($P<0.001$ and Supplementary Table 2). In *C. elegans* there is a strong correlation between decreased fat accumulation and longevity. Our observation of Nile red staining of fat was also consistent with the previous findings [25], as 4-HEG reduced fat storage in wild-type worms. Recently higher fat accumulation was found to be associated with higher ROS

production [30]. Our results also support the hypothesis that a lower level of fat storage is associated with lower ROS accumulation and hence enhanced longevity in *C. elegans*. However, we have also observed that 4-HEG supplementation did not prolonged the life span of the proaging mitochondrial mutant *mev-1(kn-1)* (Fig. 2D, Table 2), which has a short life span due to accumulation of ROS as a result of a mutation in its cytochrome *b* subunit of succinate dehydrogenase (complex II) of the electron transport chain [35]. This suggested the involvement of an endogenous ROS detoxification pathway conferring oxidative stress tolerance and reduced ROS level. Yu et al. in 2010 [42] also demonstrated similar findings in *C. elegans* treated with *Cinnamomum cassia* bark. Combined, these results show that the compound reduced the ROS level inside the wild-type worms, but it required an endogenous pathway to produce such effects.

It has also been observed that 4-HEG supplementation enhanced other age-dependent physiological parameters such as pharyngeal pumping and CI. At adult day 0, 4-HEG supplementation did not affect pharyngeal pumping of wild-type worms except at the 2 μ M concentration, which increased the pumping rates, whereas at adult day 5 it was observed that 20 μ M also increased the movement of the pharynx terminal bulb (Fig. 5A and Supplementary Table 3). The increase in pumping rates suggested the proper food intake and diminished the probability of DR-like effects. It has been already reported that *C. elegans* can modify its behavior to avoid various adverse conditions and this change in learning behavior and motor activity is associated with aging [43,44]. Consistent with previous findings [20] we also observed improved CI in adult day 5 wild-type worms (Fig. 5B), suggesting the ability of life-span-extending compounds to attenuate the age-dependent decline in physiological behaviors.

Further in this study, the genetic requirements for the longevity-promoting effects of 4-HEG in *C. elegans* were investigated. To find the possible mechanism of action of 4-HEG we examined the life span of *sir-2.1* mutants, which have been reported to have increased life span during DR; however, this is controversial [34,36,45]. The increased life span of worms lacking SIR-2.1 activity showed that SIR-2.1 is not involved in the effect mediated by 4-HEG supplementation. Further, the life span of *daf-16* mutants after 4-HEG supplementation was examined. It was observed that 4-HEG supplementation failed to increase the life span of *daf-16* mutants (Fig. 2C and Table 2). This showed the possibility of a requirement of the major longevity-regulating transcription factor DAF-16. Later, in an experiment with TJ356 mutants carrying a *daf-16::gfp* transgene, it was observed that 4-HEG treatment was able to translocate DAF-16 into the nucleus (Fig. 6C). In *C. elegans* DAF-16 was found to be a major transcription factor regulating a number of longevity- and stress tolerance-associated genes [14,46,47]. It has also been suggested that compounds restraining the IIS pathway and DAF-16 localization are provocative contenders for pharmacological manipulation of life span [47]. To evaluate the role of 4-HEG in longevity through DAF-16, a series of experiments was conducted on stress-inducible genes of *C. elegans*, viz., *hsp-16.2*, *sod-3*, and *gst-7*. 4-HEG was able to upregulate the protein level in *hsp-16.2::gfp* and *sod-3::gfp* *C. elegans* transgenic lines. Simultaneously an experiment was conducted using quantitative real-time PCR with the same genes and in addition *gst-7*. The results revealed that 4-HEG upregulated the expression of *hsp-16.2* by 1.9-fold and 1.5-fold in the case of *sod-3*, whereas *gst-7* showed no marked induction in its expression. It has been already reported that *gst-7* is regulated by the SKN-1 transcription factor [48]. Upregulation of the genes *hsp-16.2* and *sod-3* explains how 4-HEG could significantly promote the longevity and stress resistance of *C. elegans* through DAF-16. Previously other studies have also shown that increased longevity by naturally occurring molecules

required DAF-16 activation [20,38,42]. The present findings are consistent with previous studies indicating that 4-HEG supplementation results in translocation of DAF-16, leading to an increased life span and improved aging end-point parameters.

Conclusion

The present results demonstrate that 4-HEG has the potential to significantly increase the life span of worms under standard laboratory conditions of 20 °C and paraquat-induced oxidative stress as well. It was observed that 4-HEG also has the ability to attenuate ROS and fat accumulation inside worms, leading to improved physiological parameters. Finally, it was demonstrated that an endogenous ROS detoxification pathway, nuclear localization of the major transcription factor DAF-16, and upregulation of several DAF-16-regulated stress-inducible genes, viz., *hsp-16.2* and *sod-3*, are responsible for the 4-HEG-mediated longevity in *C. elegans*. This study for the first time reports the longevity-promoting activity of 4-HEG from *P. integrifolia* in the *C. elegans* model system.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.freeradbiomed.2012.08.594>.

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